Summary Two-factor ANOVA under balanced design and with replication

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LSM

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \nu_{ij} + \varepsilon_{ijk}$$

Constraints
$$\sum \alpha_i = 0 \qquad \sum_i \nu_{ij} = 0 \ \forall j$$
$$\sum \beta_j = 0 \qquad \sum_j \nu_{ij} = 0 \ \forall i$$

BLUE = LSE of $\mu, \{\alpha_i\}, \{\beta_j\}, \{\nu_{ij}\}$ are

$$\hat{\mu} = \overline{Y}_{\cdots}$$
$$\hat{\alpha}_i = \overline{Y}_{i\cdot\cdot} - \overline{Y}_{\cdots}$$
$$\hat{\beta}_j = \overline{Y}_{\cdot j\cdot} - \overline{Y}_{\cdots}$$
$$\hat{\nu}_{ij} = \overline{Y}_{ij\cdot} - \overline{Y}_{i\cdot\cdot} - \overline{Y}_{\cdot j\cdot} + \overline{Y}_{\cdots}$$

ANOVA decomposition

$$SSTO = SSA + SSB + SSAB + SSE$$
$$SSTO = \sum\sum\sum (Y_{ijk} - \overline{Y}_{\cdots})^2$$
$$SSE = \sum\sum\sum (Y_{ijk} - \overline{Y}_{ij\cdot})^2$$
$$SSA = mb \sum_{i=1}^{a} \hat{\alpha}_i^2$$
$$SSB = ma \sum_{j=1}^{b} \hat{\beta}_j^2$$
$$SSAB = m \sum_i \sum_j \hat{\nu}_{ij}^2$$

$$df(SSA) = df(\underline{\hat{\alpha}}) = a-1$$
$$df(SSB) = df(\underline{\hat{\beta}}) = b-1$$
$$df(SSAB) = df(\underline{\hat{\nu}}) = ab - (a+b-1) = (a-1)(b-1)$$

Suppose $\varepsilon_{ijk} \overset{iid}{\sim} N(0, \sigma^2)$

$$\mathbb{E}(SSA) = mb \sum \alpha_i^2 + (a-1)\sigma^2$$
$$\mathbb{E}(SSB) = ma \sum \beta_j^2 + (b-1)\sigma^2$$
$$\mathbb{E}(SSAB) = m \sum\sum \nu_{ij}^2 + (a-1)(b-1)\sigma^2$$
$$\mathbb{E}(SSE) = \underbrace{(m-1)ab}_{\to\, df(SSE)}\sigma^2$$

ANOVA decomposition

$$\sum_i \sum_j \sum_k Y_{ijk}^2 = mab\,\overline{Y}_{\cdots}^{\,2} + SSA + SSB + SSAB + SSE$$

Fisher-Cochran $\underset{\sim}{Y} \sim N(\underset{\sim}{\theta}, I)$

$Y^T Y = \sum_{j=1}^{r} Y^T A_j Y$ where A_j's are nonnegative definite

and $\sum_{j=1}^{r} \text{rank}(A_j) = n$

$n = mab$

$\text{rank}(A) = a-1$ $\quad Y = \begin{pmatrix} Y_{111} \\ \vdots \\ Y_{abm} \end{pmatrix}$

$Y \longmapsto \hat{\alpha} = AY$

$SSA = mb \sum \hat{\alpha}_i^2$
$\quad = mb\, Y^T A^T A Y$

$\hat{\beta} = BY \hookrightarrow \text{rank}(B) = b-1$

$\hat{\gamma} = CY \hookrightarrow \text{rank}(C) = (a-1)(b-1)$

Using Fisher-Cochran, we claim that

1. $\overline{Y}_{\cdots}$, SSA, SSB, SSAB, SSE are indep.
2. $\frac{SSA}{\sigma^2} \sim \chi^2_{a-1}\left(\frac{\delta_A}{2}\right)$

   $\frac{SSB}{\sigma^2} \sim \chi^2_{b-1}\left(\frac{\delta_B}{2}\right)$

   $\frac{SSAB}{\sigma^2} \sim \chi^2_{(a-1)(b-1)}\left(\frac{\delta_C}{2}\right)$

   $\frac{SSE}{\sigma^2} \sim \chi^2_{(m-1)ab}$

Check that $\delta_A = \frac{mb \sum \alpha_i^2}{\sigma^2}$ $\quad \delta_C = \frac{m \sum_i \sum_j \gamma_{ij}^2}{\sigma^2}$

$\delta_B = \frac{ma \sum \beta_j^2}{\sigma^2}$

ANOVA Table

Source of Variation	df	SS	MS	F-statistic
Factor A main effects	$a-1$	SSA	$SSA/(a-1)$	$F_A = \frac{MSA}{MSE}$
Factor B main effects	$b-1$	SSB	$SSB/(b-1)$	$F_B = \frac{MSB}{MSE}$
AB interaction effects	$(a-1)(b-1)$	SSAB	$SSAB/(a-1)(b-1)$	$F_{AB} = \frac{MSAB}{MSE}$
Error	$(m-1)ab$	SSE	$SSE/(m-1)ab$	
Total	$mab-1$	SSTo		

Consider $H_0^A : \alpha_1 = \dots = \alpha_a = 0$ vs. H_1^A : not all α_i's are zero

(test for presence of main effects of factor A)

Full Model: $Y_{ijk} = \mu + \alpha_i + \beta_j + \nu_{ij} + \varepsilon_{ijk}$

Reduced Model$_{(A)}$: $Y_{ijk} = \mu + \beta_j + \nu_{ij} + \varepsilon_{ijk}$

together with identifiability constraints

Q) LSE of $(\mu, \underset{\sim}{\beta}, \underset{\sim}{\nu})$ remains unchanged under the reduced model.

Recall:

$$G(\mu, \underset{\sim}{\alpha}, \underset{\sim}{\beta}, \underset{\sim}{\nu}) = \sum_i \sum_j \sum_k (Y_{ijk} - \mu - \alpha_i - \beta_j - \nu_{ij})^2$$

$$\underset{\text{under } *}{=} \sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{ij\cdot})^2 + mb \sum_i (\hat{\alpha}_i - \alpha_i)^2$$

$$+ ma\sum_j (\hat{\beta}_j-\beta_j)^2 + m\sum_i\sum_j(\hat{\gamma}_{ij}-\gamma_{ij})^2$$

$$+ mab(\hat{\mu}-\mu)^2$$

$$SSE^{(A)}_{red} = \sum_i\sum_j\sum_k \left(Y_{ijk}-\hat{\mu}-\hat{\beta}_j-\hat{\gamma}_{ij}\right)^2$$

$$= \underbrace{SSE}_{full} + SSA$$

$$am\left\{\begin{pmatrix} \left.\begin{matrix}Y_{111}\\ \vdots \\ Y_{11m}\end{matrix}\right] \to i=1, j=1 \\ \left.\begin{matrix}Y_{211}\\ \vdots\\ Y_{21m}\end{matrix}\right] \to i=2, j=1 \\ \vdots\\ Y_{a11}\\ \vdots\\ Y_{a1m}\\ \hline \vdots\end{pmatrix}\right.
=
\begin{array}{c} \\ \left(\begin{array}{c|cccc|ccc|cccc} 1_m & 1_m & 0 & & 0 & 1_n & & & 1_m & 0 & & \\ 1_m & 0 & 1_n & & 0 & 1_n & 0 & & 0 & 1_n & & \\ \vdots & \vdots & \vdots & & \vdots & \vdots & & & & & \ddots & \\ 1_m & 0 & 0 & & 1_n & 1_n & & \cdots & & & & \\ \hline 1_m & 1_n & 0 & & 0 & 0 & 1_n & & & & & \\ \vdots & 0 & 1_n & & 0 & & 1_n & & & & & \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & & & & \end{array}\right) \end{array}
\begin{pmatrix}\mu\\ \underline{\alpha}\\ \underline{\beta}\\ \underline{\gamma}\end{pmatrix}$$

with the column blocks labelled X_A, X_B, X_{AB} and the first a rows braced by a.

$$\underline{Y} = \mu 1_n + X_A\underline{\alpha} + X_B\underline{\beta} + X_{AB}\underline{\gamma} + \varepsilon$$

$1_n \in \mathcal{C}(X_A) \qquad 1_n \in \mathcal{C}(X_B)$

$\mathcal{C}(X_A) \subset \mathcal{C}(X_{AB}) \quad \mathcal{C}(X_B) \subset \mathcal{C}(X_{AB})$

X_{AB} has orthogonal columns

Kronecker product of matrices

Let $A_{m\times n}$ and $B_{p\times q}$

Define $A\otimes B$ to be the $mp\times nq$ matrix

$$\begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ & \vdots & & \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{bmatrix}$$

$$A\otimes B \neq B\otimes A$$

What is $A\otimes B\otimes C$

$$A\otimes B\otimes C \overset{\text{def}}{=} (A\otimes B)\otimes C \overset{\text{check}}{=\!=} A\otimes(B\otimes C)$$

(i.e. $\otimes$ is associative)

$$A\otimes B + D\otimes B$$
$$= (A+D)\otimes B$$

$$(A\otimes B)(C\otimes D) = (AC)\otimes(BD)$$

$$1_n = 1_{ab}\otimes 1_m = 1_a\otimes 1_b\otimes 1_m$$
$$X_{AB} = I_{ab}\otimes 1_m = I_a\otimes I_b\otimes 1_m$$

Use the Y_{ijk} where we change index k first then change index j and change index i, while stacking.

$$\begin{pmatrix} Y_{111} \\ \vdots \\ Y_{11m} \\ Y_{121} \\ \vdots \\ Y_{12m} \\ \vdots \\ Y_{1bm} \\ \hline \vdots \end{pmatrix} = \begin{array}{c} \begin{array}{cccc} 1_n & X_A & X_B & X_{AB} \end{array} \\ \left(\begin{array}{c|cc|ccc|c} & 1_n & & 1_m & & 0 & \\ \vdots & \vdots & 0 & & \ddots & & \vdots \\ & 1_n & & 0 & & 1_n & \\ \hline & & 1_n & 1_m & & 0 & \\ & 0 & \vdots & & \ddots & & \\ & & 1_n & 0 & & 1_m & \\ & & \vdots & & \vdots & & \end{array}\right) \end{array}$$

$$X_A = I_a \otimes 1_{bm} = I_a \otimes 1_b \otimes 1_m$$

$$X_B = 1_a \otimes I_b \otimes 1_m$$

Further results for Kronecker products

→ $(A \otimes B)^T = A^T \otimes B^T$

→ If A and B are non-singular

$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$

$\left(\because (A \otimes B)(A^{-1} \otimes B^{-1}) = (AA^{-1} \otimes BB^{-1}) = I_n \times I_p = I_{np}\right)$

→ If A^- and B^- are g inverse of A and B resp.,

then $A^- \otimes B^-$ is a g inverse of $A \otimes B$

→ $P_{A \otimes B} = P_A \otimes P_B$

orthogonal projector

$$\| $$

$$(A \otimes B)\left((A \otimes B)^T (A \otimes B)\right)^- (A \otimes B)^T$$

$$= (A \otimes B)(A^TA \otimes B^TB)^- (A \otimes B)^T$$

by choice

$$= (A \otimes B)\left((A^TA)^- \otimes (B^TB)^-\right)(A^T \otimes B^T)$$

$$= \left(A(A^TA)^- A^T\right) \otimes \left(B(B^TB)^- B^T\right)$$

$$P_{1_n} = P_{1_a} \otimes P_{1_b} \otimes P_{1_m}$$

$$P_{X_A} = P_{I_a} \otimes P_{1_b} \otimes P_{1_m} = I_a \otimes P_{1_b} \otimes P_{1_m}$$

$$P_{X_B} = P_{1_a} \otimes I_b \otimes P_{1_m}$$

$$P_{X_{AB}} = I_a \otimes I_b \otimes P_{1_m}$$